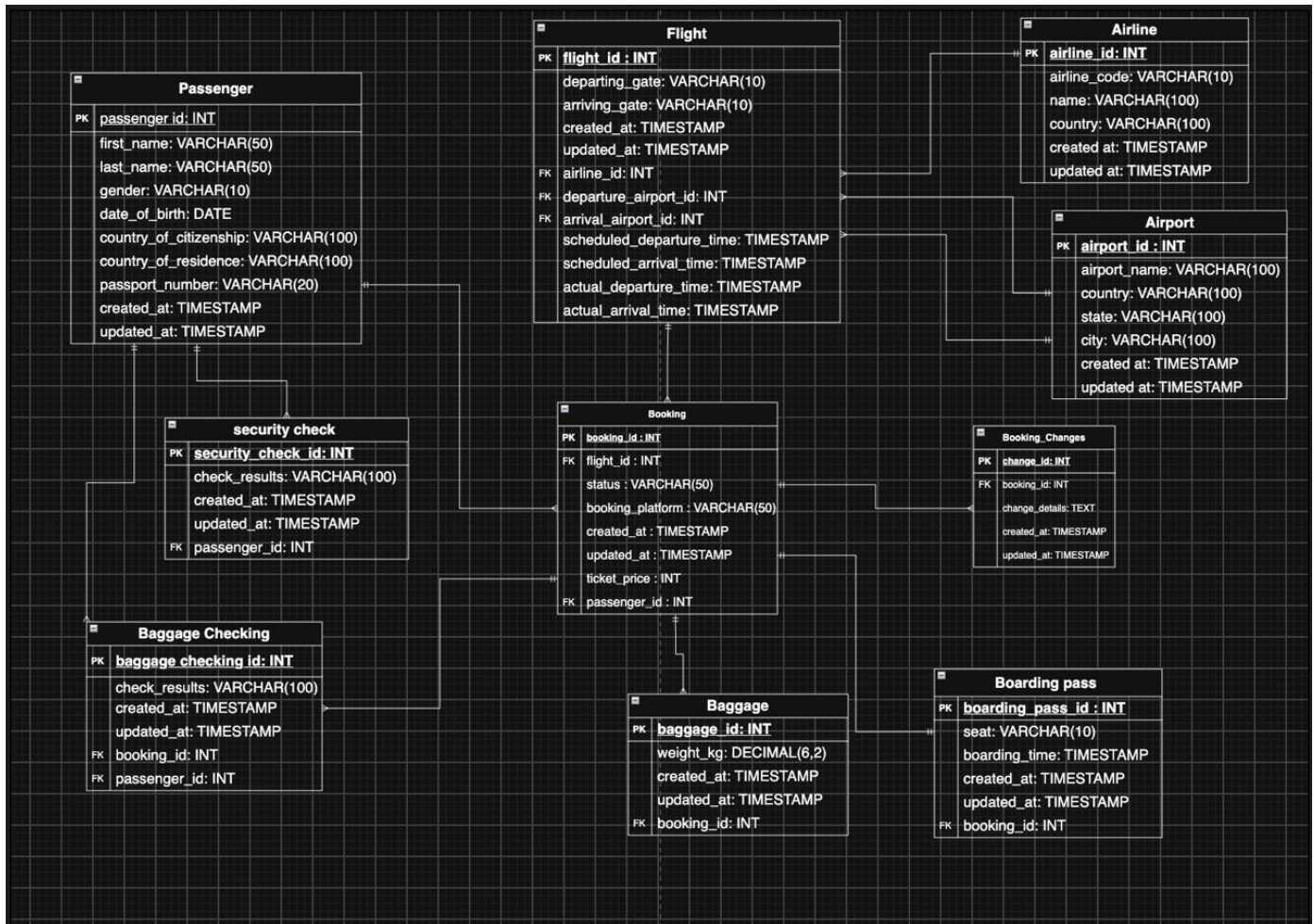


Laboratory Work 1

ERD Diagram: International Airport Database

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Entity-Relationship Diagram of the International Airport Database.

1. Legend

Symbol / notation	Meaning
PK	Primary Key -uniquely identifies a record in an entity
FK	Foreign Key - references a primary key in another entity
1:N	One-to-Many relationship
1:1	One-to-One relationship

2. Entity Descriptions

Entity	Description
Airport	Stores information about airports: ID, name, country, state, city, creation date, and update date.
Flight	Stores scheduled and actual flight information, departure and arrival gates and times, airline ID, and departure and arrival airport IDs.
Airline	Stores airline information including ID, code, name, country, creation date, and update date.
Booking	Stores passenger flight bookings, including booking ID, flight ID, status, booking platform, timestamps, ticket price, and passenger ID.
Boarding Pass	Stores boarding pass information including ID, seat, boarding time, timestamps, and booking ID.
Baggage	Stores registered baggage information including ID, weight in kilograms, dates, and booking ID.
Baggage Checking	Stores baggage checking information including checking ID, results, timestamps, booking ID, and passenger ID.
Passenger	Stores passenger profile information including ID, name, gender, date of birth, citizenship, residence, passport number, and timestamps.
Security Check	Stores airport security check information including ID, results, timestamps, and passenger ID.
Booking Changes	Stores changes made to bookings separately, including change ID, booking ID, change details, and timestamps.

3. Relationship and Cardinality

Relationship	Description
Airline - Flight	One-to-Many (1:N). A flight belongs to an airline through <code>Flight.airline_id</code> .
Airport - Flight (Departure)	One-to-Many (1:N). A flight references its departure airport through <code>Flight.departure_airport_id</code> .
Airport - Flight (Arrival)	One-to-Many (1:N). A flight references its arrival airport through <code>Flight.arrival_airport_id</code> .
Flight - Booking	One-to-Many (1:N). A booking references the flight through <code>Booking.flight_id</code> .
Passenger - Booking	One-to-Many (1:N). A booking references the passenger through <code>Booking.passenger_id</code> .
Booking - Boarding Pass	One-to-One (1:1) in this simplified model. The boarding pass references its booking through <code>booking_id</code> .
Booking - Baggage	One-to-Many (1:N). Registered baggage is associated with a booking through <code>Baggage.booking_id</code> .
Booking - Baggage Checking	One-to-Many (1:N). Baggage checking records are associated with a booking through <code>Baggage_Checking.booking_id</code> .
Passenger - Baggage Checking	One-to-Many (1:N). Baggage checking records reference the passenger through <code>Baggage_Checking.passenger_id</code> .
Passenger - Security Check	One-to-Many (1:N). Security check records reference the passenger through <code>Security_Check.passenger_id</code> .
Booking - Booking Changes	One-to-Many (1:N). A booking can have multiple recorded changes; each change references the booking through <code>booking_id</code> .

Conclusion

In conclusion, the ERD shows the main parts of the International Airport database. It includes airports, flights, airlines, passengers, bookings, boarding passes, baggage, and security checks. The relationships between these entities show how the data is connected. Primary keys and foreign keys help keep the data correct and organized. This ERD gives a clear structure for managing airport information.